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Efficacy of NER Models on Different Domains

AGENDA

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03 CORE PROBLEM

04 DATASETS

05 RESULTS

06 NEXT STEPS

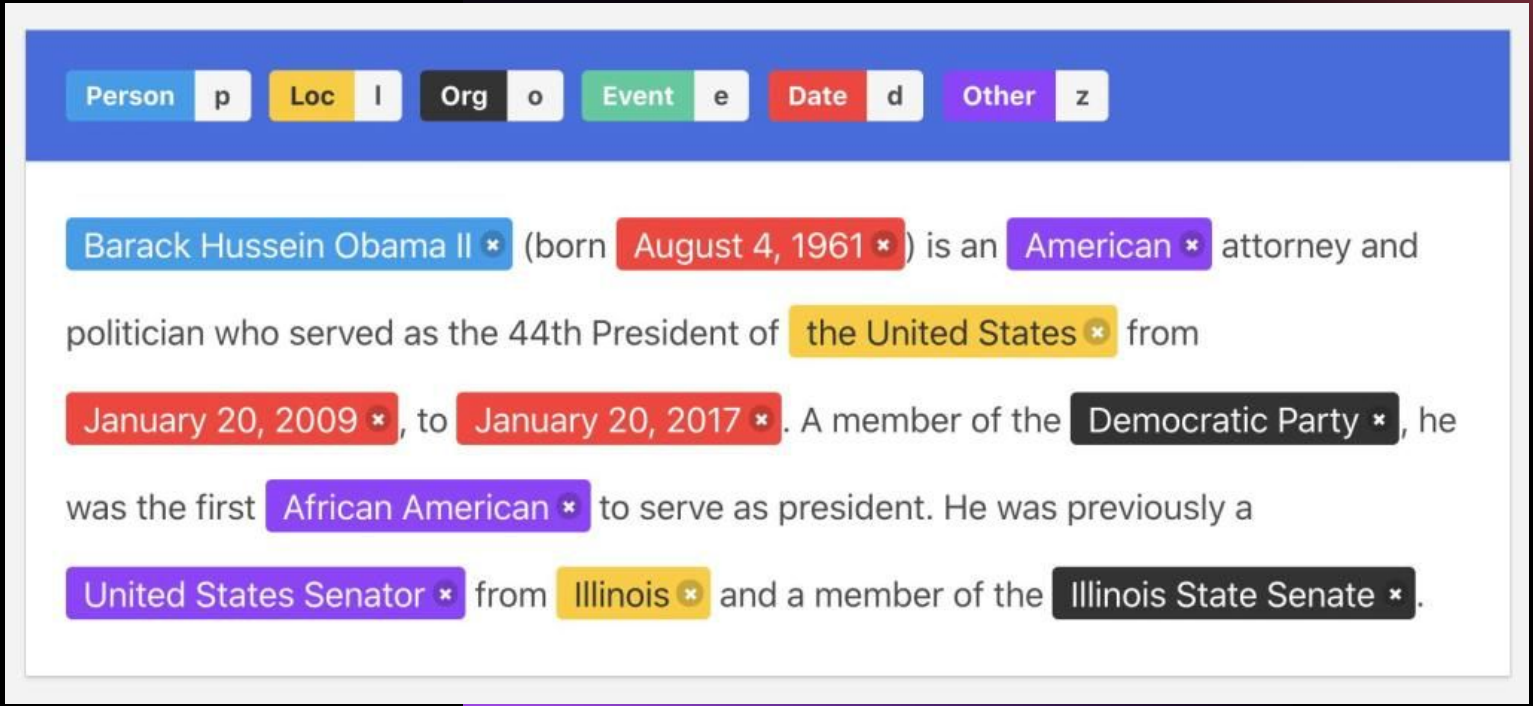
Leading Question

How does a Named Entity Recognition (NER) model's performance differ when applied to fiction versus nonfiction text, and which entity types are most sensitive to differences between these genres?

2. BACKGROUND

- Named Entity Recognition (NER): An NLP task that identifies and labels entities such as
  - PER[illegible]cation), ORG (Organiz[illegible]
- NER is widely used in:
  - Search & inform[illegible]
  - Question answeri[illegible]
  - Content classificatio[illegible] graph construction
- Many NER models are trained heavily on newswire or nonfiction datasets
- Fiction introduces challenges:
  - Creative language, idioms,[illegible]
  - Unconventional names and[illegible]
  - Entities without explicit cont[illegible]

NER Example Graphic



2. BACKGROUND
Relation to prior work

- Wang et al., 2020 studies how models degrade when transferred to another domain it was not trained on.
- Performance changes significantly on
  - Person and [illegible]nes
  - Less standardiz[illegible]nd in fiction
- Our project builds on this by
  - Directly comparing fic[illegible]
  - Identifying which entity [illegible] affected

Wang et al., 2020: Multi-Domain Named Entity Recognition with Genre-Aware and Agnostic Inference

2. BACKGROUND
Relation to prior work

Bamman et al., 2019: An Annotated Dataset of Literary Entities

- Highlights challenges for NER in fiction:
  - Personal
  - Metaphor
- Larger proportion of PER and LOC entities in literary text

Train → Test	Precision	Recall	F
ACE → ACE	75.3 [72.7–77.9]	63.3 [60.5–66.1]	68.8 [66.2–71.2]
ACE → Literature	57.8 [54.2–61.5]	37.7 [35.1–40.6]	45.7 [42.8–48.6]
Literature → Literature	75.1 [72.1–77.8]	62.6 [59.7–65.4]	68.3 [65.5–70.8]

3. CORE PROBLEM

- NER models struggle to generalize from nonfiction to fiction
- Performance inconsistencies vary by entity type
  - PER appear[illegible]ction, but with unusual naming[illegible]
  - ORG and LOC ca[illegible] standardized or ficti[illegible]
- Understanding where performance breaks down helps:
  - Improve dataset diversity[illegible]
  - Better evaluate models in[illegible] domains
  - Guide targeted model retra[illegible] augmentation

4. DATA USED

- LitBank - Fiction
  -
  - 100 w[illegible] language fiction
- OntoNotes 5.0 - Nonfiction
  - ~2 million toke[illegible]
  - 3,600 English Ne[illegible] Broadcast docume[illegible]

4. DATA USED

- Converted original annotated labels into a simplified tag set
  - **Person** (PER)
  - **Location** (LOC)
  - **Organization** (ORG)
  - **Other / Not an** (O)
- Reduced complexity
- Helped isolate performance differences by entity type

5. RESULTS (PHASE 1)

- Model: **distilbert-NER**
  - [illegible] Face)
  - Fine-[illegible] 2003 NER dataset ([illegible]
  - Recognizes [illegible] and ORG entities
- Evaluate on separate test sets for fiction and nonfiction data
  - LitBank and OntoNo[illegible]

5. RESULTS (PHASE 1)

LitBank (fiction)

	precision	recall	f1-score	support
LOC	0.02	0.01	0.01	227
MISC	0.00	0.00	0.00	0
ORG	0.08	0.10	0.09	20
PER	0.11	0.03	0.05	1502
micro avg	0.07	0.03	0.04	1749
macro avg	0.05	0.04	0.04	1749
weighted avg	0.09	0.03	0.05	1749

OntoNotes (nonfiction)

	precision	recall	f1-score	support
LOC	0.04	0.61	0.08	179
MISC	0.00	0.00	0.00	0
ORG	0.54	0.61	0.58	1795
PER	0.82	0.72	0.77	2337
micro avg	0.35	0.67	0.46	4311
macro avg	0.35	0.49	0.36	4311
weighted avg	0.67	0.67	0.66	4311

6. NEXT STEPS

- Fine tune a distilbert-NER model
  - [illegible] ing set of fiction and [illegible]
  - To impro[illegible] tion across writing styles [illegible]
- Re-Evaluate Performance
  - Analyze which e[illegible] from combined trai[illegible]

THANK YOU

Any questions?